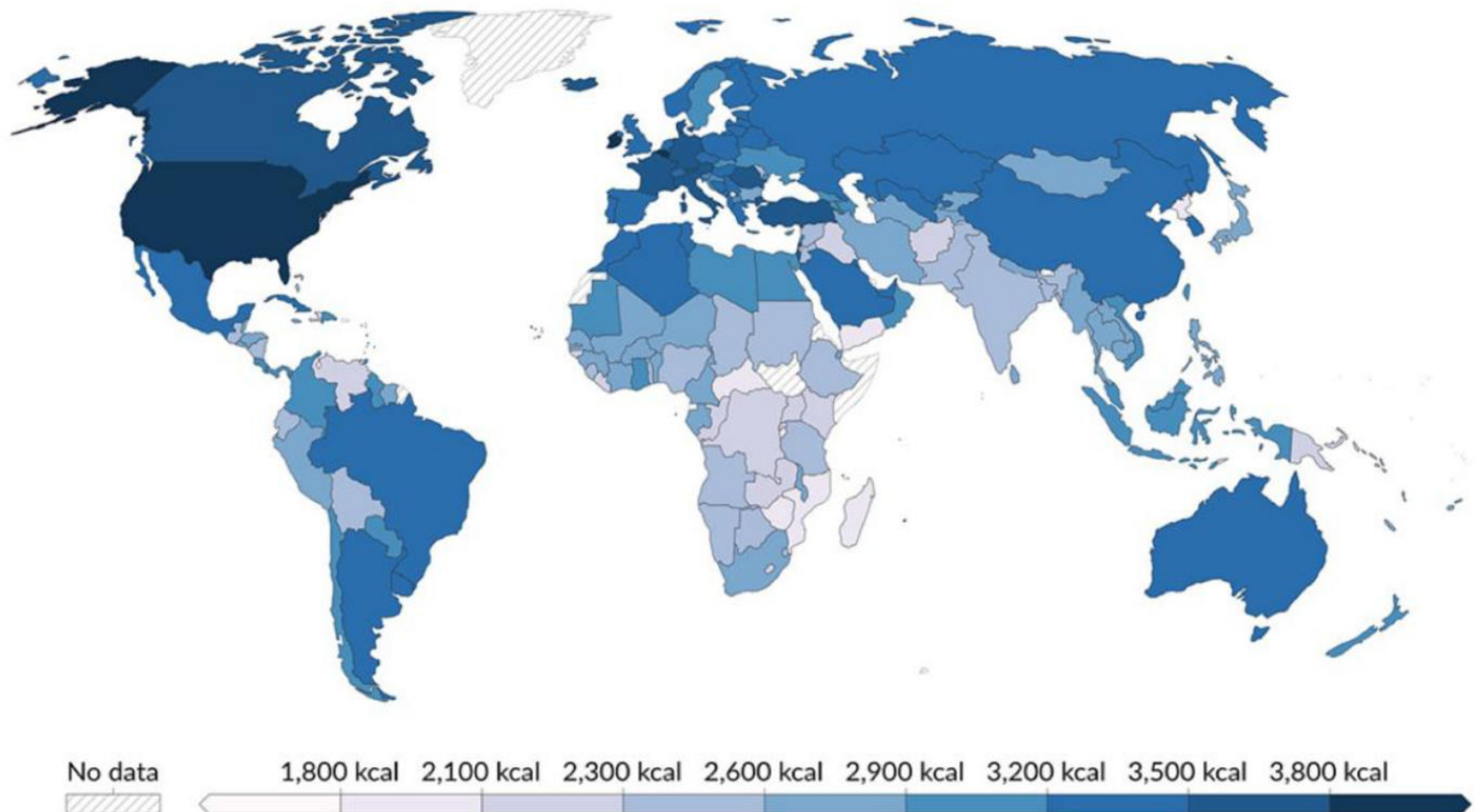


10.2 Global patterns of food supply and demand

Globally, **calorie intake** varies significantly both between countries and within countries (Source A). The main cause for an increase in calorie intake is economic development. The daily supply of calories as protein and fat all tend to increase with increased wealth. People in low-income countries, such as Ethiopia, may eat less than 2100 calories a day, with their diet being predominantly plant-based, and consisting mainly of grains and roots. In comparison, people in high income countries, such as the USA, may eat as many as 3600 calories per day, predominantly as meats, saturated fats, and processed grains with added sugars.

There are a number of reasons for the increase in demand for food globally; foremost is the ongoing increase of the human population. In addition, economic development results in people having more money to spend, so they are able to purchase more food, increasing the amount and diversity of food they are able to access. Furthermore, as a country's economy strengthens, more money is available for investment into food production and storage, or the import of food. This increases the availability of a more diverse diet to the population and a higher calorie intake per person over time.



Source **A**: Daily supply of calories per person (2018)

TASK 1: Study Source **A**

- Compare the distribution** of countries with a high-calorie intake to those with a low-calorie intake.
- Explain the pattern of calorie intake that you identified in part a.
- In pairs, carry out research about the calorie intake for your country and then your town. Are there areas that have a lower or higher calorie intake per person?
- Draw a sketch map** of your country and look for patterns of low or high calorie intake per person.

REFLECTION

Did drawing a sketch map help you to identify patterns more easily? Are there other topics where you can draw sketch maps to improve your understanding of the content?

The world has experienced a shift in food systems and diets, with both food trade and agricultural trade expanding rapidly in the 2000s. The access, availability, and quality of food is changing, with the need for proper nutrition and food security becoming a global discussion.

The causes of the **globalisation of food supplies** are complex. However, there are a few underlying factors:

Economic causes

Economic development is central to the globalisation of food supplies. As economies develop, so too do transport routes. This supports the ongoing export and import of food products between countries. The mechanisation of transport and the ability to freeze or preserve food products has also resulted in moving foods from where they were produced to a distant market quickly and efficiently. In addition, urbanisation and foreign direct investment in food production has led to greater availability, affordability, and consumption of processed foods, which have a longer shelf life and are easier to transport.

Social causes

Social factors, such as an increased number of women in the workforce and higher rates of rural to urban migration, have increased the need for commercial farming. Commercial farming results in an increase in food production, which leads to surplus food. This surplus food allows food companies to sell to areas where farming is not occurring (e.g. cities), outside of the production region or country. In addition, increased migration globally results in a demand for food products from ‘back home’. This increases the movement of food globally.

Large food companies use social media and other marketing methods to increase awareness around the world to the different types of food products available. This results in an increase in demand for food types that are produced far from where they are consumed.

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Environmental causes

Underlying the movement of food products around the world is the environment. Different plants and animals thrive in different environments. This means that a food product that a population enjoys might not be grown locally. This drives the movement of food between locations around the planet.

The globalisation of food supplies has social, economic, and environmental impacts. Some positive impacts are the availability of a wider variety of food products around the globe and the expansion of international trade. However, there are a number of negative impacts.

Economic impacts

Transnational food corporations put the small local farmer at a disadvantage. Local farmers are unable to compete with the marketing capabilities and the lower price that large companies can sell food at. As a result, food security, local food production, and the presence of healthier, home-grown food can reduce as the small local farmer is put out of business.

Local economies are affected by international changes in food supply or food prices, impacting people who are already unable to afford their families. This leads to increased **food insecurity**, which is also a social impact.

Social impacts

The number of malnourished people is increasing. This is due to nutritional deficiencies and dietary imbalances, and not necessarily just food shortages. The globalisation of food has increased access to cheap fast-foods with low nutritional value. Eating ultra-processed foods increases the risk of health challenges like high blood pressure, strokes, and heart attacks by as much as 24%.

Environmental impacts

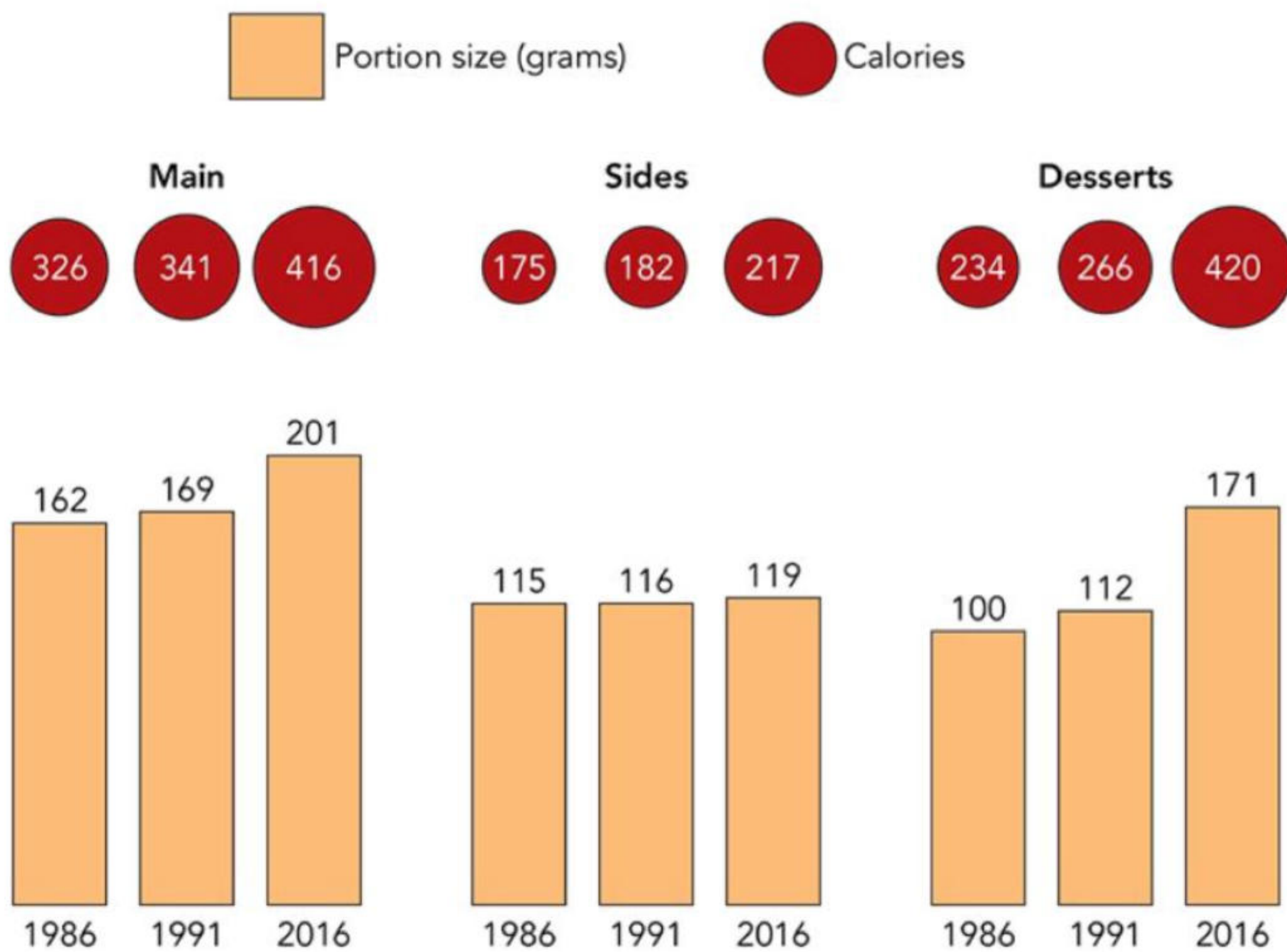
Farmers often use modern industrial agricultural techniques, sacrificing long-term sustainability of the farmland for higher productivity. Soils become exhausted, no longer able to support plant growth and erosion. In the long term, this becomes both an economic and social impact as crops fail, and both hunger and poverty increase.

To meet global demand and speed up production, commercial farmers often use fossil fuels, chemical fertilisers, and pesticides, which pollute the environment. For ease of planting and harvesting, commercial farmers plant the same crop over large areas year after year in a process known as **monocropping**. This results in decreased biodiversity as a single species replaces the diverse ecosystem that existed in the area before.

In addition, food is transported longer distances to the market, increasing its ‘food miles’. Food miles are the distance food is transported during the journey from producer to consumer, as a measurement of the amount of fossil fuel used to transport it. The greater the food miles, the greater the amount of carbon emitted to produce and transport the food. This carbon adds to the greenhouse gases in the atmosphere, increasing the risk of climate change.

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Calorie count and size of portions at 10 US fast food chains*



* Chains: Arby's, Burger King, Carl's Jr, Dairy Queen, Hardee's, Jack in the Box, KFC, Long John Silver's, McDonald's and Wendy's

Source B: The change in portion sizes and calorie value of fast food between 1980 and 2016. The average adult human requires 2000 calories (female) and 2500 calories (male) per day.

TASK 2: Study Source B

- Calculate the increase** in calories for desserts between 1986 and 2016.
- Explain why an increase in the size of fast-food portions would be considered unhealthy.
- In pairs, discuss the marketing of fast food and how it impacts fast-food sales and the economy of a country, region, or town.

Strategies for increasing food supplies

An increase in human populations, in combination with a decrease in healthy agricultural land area, has resulted in the development of techniques to increase food output from farms. The Green Revolution was a period of agricultural development that started in the early 20th century and continued through until the late 1980s. During this period, improvements in irrigation, chemical controls, fertilisers, **high-yield variety (HYV)**, and **genetically modified (GM) crops**, and mechanisation all increased farming outputs (Source C).

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Source C: Strategies to increased crop yields

High-yielding variety (HYV) and genetically modified (GM) crops: Crops that have been bred or produced to increase the speed of growth and yield. This increases the output from a farm, increasing food production.

Fertilisers: natural or synthetic products added to soils to supply plants with nutrients, such as nitrogen. This allows for better and faster crop growth and produces more food per plant.

Irrigation: The supply of water to land or crops using mechanised systems. This allows for larger areas and areas further from water sources to be farmed, increasing the potential output from farms.

Strategies to increase food supplies

Herbicides and pesticides: Chemical controls for insects and plants that reduce the productivity of plants. By adding these chemicals, the number of pests or competing plants is controlled, increasing crop harvests.

Mechanisation: The use of machines to plough, plant, harvest and process crops. This increases the area that can be farmed as it is faster than human labour. It also helps to farm in soils that are harder to work and increases the outputs from farm land.

► Image Description

Despite the benefits of increased food production, these strategies have introduced a number of negative impacts:

- Mechanisation resulted in the loss of jobs for many farm workers, forcing them to move to new areas to look for work.
- Herbicides, pesticides, and fertilisers all have negative impacts on the environment if they are not carefully managed. They can cause decreased biodiversity, and contaminate soils and waterways.
- High-yield crops (HYVs and GM crops) grow faster and require more fertiliser and water. Farmers who do not have the money to invest in the machines and resources required to farm this type of crop are outcompeted by those who are able to.
- The increased use of water and natural areas for farming puts pressure on ecosystems.

TASK 3: Study Source C

- Write a 200 to 250-word essay on how the strategies outlined in Source C have increased food output from farms.
- In pairs, research the main problems that the techniques used to increase food output have caused. Create a spider diagram to show your findings. Consider the following factors in your research:
 - the impact of the increased use of herbicides and pesticides
 - the impact of the increased use of fertilisers
 - the clearing of natural areas for new farmland
 - the overuse or misuse of water resources
 - the mismanagement of soils (soil exhaustion/erosion).